Vinay Babu Gorantla

Generative AI Engineer | RAG | Agentic AI | LLM Applications | MLOps

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Portfolio: Vinay Babu Gorantla - Portfolio

Professional Summary:

Generative AI Engineer (RAG & Agentic Systems) with 7+ years of experience designing and delivering enterprise-grade AI applications. Specialized in building RAG pipelines, LLM-powered assistants, and Agentic AI workflows using Azure OpenAI, LangChain, and LangGraph, with strong focus on accuracy, observability, and scalability.

Proven ability to design end-to-end AI architectures integrating retrieval, reasoning, and workflow orchestration, delivering measurable impact through automation and decision intelligence. Strong foundation in Machine Learning and MLOps, with expertise in deploying production-ready systems using MLflow, Docker, CI/CD, and Azure cloud platforms.

Experience Summary:

- Designed and deployed Generative AI applications (RAG + Agentic workflows), reducing manual document search by 60% and analysis effort by 50%.
- Built hybrid retrieval systems using Azure AI Search and vector databases, improving retrieval relevance by 25%.
- Implemented LLM evaluation and observability frameworks (LangSmith, RAGAS), improving response accuracy by 30%.
- Delivered scalable AI systems and automation pipelines, accelerating business decision-making and operational efficiency.
- Developed and productionized ML models and data pipelines, improving prediction accuracy by 20–25% and reducing data processing time by 40%.
- Established MLOps best practices and CI/CD pipelines, reducing deployment effort by 50–60%.

Previous Employment History:

Machine Learning Engineer (Contract)| VM2R Services LTD, City of Bristol, England United Kingdom

| Feb 2022 to Present

- Designed and developed RAG-based document intelligence platforms using Azure OpenAI, LangChain, and LangGraph, reducing document search time by 60%.
- Built agentic AI workflows for summarization, reasoning, and decision support, reducing manual analysis effort by 50%.
- Implemented hybrid retrieval architectures using Azure AI Search, FAISS, and ChromaDB, improving retrieval relevance by 25%.
- Established evaluation and monitoring pipelines using LangSmith and RAGAS, improving response accuracy by 30% and reducing hallucinations.
- Developed LLM-powered assistants and workflow automation systems, improving operational efficiency and decision-making speed.
- Designed and deployed end-to-end ML pipelines (classification, regression, anomaly detection, time-series forecasting), improving model performance by up to 20%.
- Built scalable data ingestion and feature engineering pipelines, reducing processing time by 40%.
- Productionized ML systems using MLflow, DVC, Docker, and FastAPI, reducing deployment time by 50%.
- Implemented CI/CD pipelines with Azure DevOps and GitHub Actions, reducing deployment effort by 60%.
- Optimized ETL workflows using Azure Data Factory, Databricks, and Synapse, improving reliability and reducing failures by 30%.

- Collaborated with cross-functional teams to deliver AI solutions, improving turnaround time for insights.

Data Scientist | WIPRO Technologies ltd, Bangalore, India | Nov 2018 to Jan 2022

- Delivered end-to-end data science solutions, improving decision-making speed by 35%.
- Developed regression and forecasting models, improving prediction accuracy by 15–25%.
- Built and optimized data pipelines, reducing data preparation time by 40%.
- Conducted EDA and statistical analysis, identifying key drivers that improved operational efficiency by 20%.
- Identified inefficiencies in banking processes, contributing to 25% improvement in process efficiency.
- Built Power BI dashboards, reducing reporting time by 50% and enabling real-time insights.
- Automated workflows using Python and SQL, reducing manual effort by 60%.
- Worked with stakeholders to deploy analytics solutions into production systems.

Software Engineer | HCL Technologies, Chennai, India | Jun 2016 to Nov 2018

- Managed IT service management processes (Incident, Change, Problem Management) using BMC Remedy.
- Diagnosed and resolved recurring issues, improving system stability and reducing incident frequency.
- Supported deployment and release cycles across environments.
- Wrote and optimized complex SQL queries for diagnostics and reporting.
- Created documentation and conducted knowledge transfer sessions.

Key Projects

RAG-based Document Intelligence System

- Built enterprise search and Q&A system using Azure OpenAI + LangChain + Azure AI Search
- Achieved 60% reduction in document search time
- Implemented hybrid retrieval and evaluation pipelines

Agentic Workflow Automation System

- Designed multi-step AI agents for summarization, reasoning, and decision support
- Reduced manual analysis effort by 50%
- Integrated with enterprise workflows for real-time automation

Educational Qualifications:

B. Tech Computer Science and Engineering | VIGNAN's University | Aug 2010 to Apr 2014

Technical Skills Summary:

Generative AI & LLMs

LangChain, LangGraph, LangSmith, LangServe
Azure OpenAI, Azure AI Foundry, OpenAI API, Hugging Face, Claude
RAG Pipelines, Prompt Engineering, Agentic AI Workflows
Vector DBs: FAISS, ChromaDB, Pinecone, Weaviate, Azure AI Search

MLOps & Deployment

MLflow, DVC, Docker, FastAPI
CI/CD: Azure DevOps, GitHub Actions
Model Monitoring, Experiment Tracking

Machine Learning & NLP

Scikit-learn, XGBoost, LightGBM, TensorFlow, Keras
NLP, Time-Series Forecasting, Anomaly Detection

Data Engineering

Pandas, NumPy, SQL, PySpark
ETL Pipelines, Feature Engineering

Cloud & Tools

Azure (Azure ML, Databricks, Synapse, Data Factory)
AWS (Lambda, S3, Step Functions – hands-on)
Git, Jupyter, VS Code

Programming

Python (Advanced), SQL, Bash